

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 8: Databases

Subtopic 8.1: Database Concepts

Concept 8.1.4: Normalisation (1NF, 2NF, 3NF)

Total Questions: 15

Mark Schemes begin on Page 13

Question 1 | Source: 9618_w25_qp_11 (Q2.b)

- 2 A relational database, `WORKEXP`ERIENCE, stores data about students and the companies where they complete work experience.

Students can complete multiple work experience placements but can only complete one placement at a time.

Students can complete more than one placement at the same company.

Part of the database is shown:

`STUDENT`(StudentID, FirstName, TelephoneNumber, UniversityName)

`PLACEMENT`(PlacementID, StudentID, CompanyID, StartDate, EndDate, Complete)

`COMPANY`(CompanyID, CompanyName, MaxStudentsPerPlacement)

- (b) Explain why the database `WORKEXP`ERIENCE is in Third Normal Form (3NF).

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Question 2 | Source: 9618_s25_qp_11 (Q5.d)

- 5 A company builds and sells furniture to customers. The company stores data about customers, their payment cards and their furniture orders in a database.

The database, `FURNITURE`, has the following tables:

`CUSTOMER`(CustomerID, Name, Phone)

`CUSTOMER_CARD_DATA`(CardID, CustomerID, CardType, CardNumber, EndDate)

`ORDER`(OrderID, CustomerID, TotalCost, Paid, OrderDate, Complete)

`ORDER_ITEM`(OrderItemID, OrderID, Type, Height, Width, Depth, Details)

The primary keys are underlined in each table.

The attribute `Complete` in the table `ORDER` stores the Boolean value `TRUE` if the order has been built and `FALSE` if the order has not been built.

- (d) Explain the reasons why the data in the table `ORDER_ITEM` cannot be stored in the table `ORDER`

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Question 3 | Source: 9618_s25_qp_12 (Q5.a)

- 5 An online game has a database that stores data about users and the characters each user creates in the game. Each user can create multiple characters and purchase multiple items for each character.

The normalised database has the following design:

`USER(Username, Password, DateOfBirth)`

`CHARACTER(CharacterName, CharacterID, Username, Level, Money)`

`ITEM(ItemName, MinimumLevel, Cost)`

`CHARACTER_ITEM(CharacterID, ItemName)`

- (a) Explain the purpose of the table `CHARACTER_ITEM` in the database.

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Question 4 | Source: 9618_w24_qp_13 (Q4.c.ii)

- 4 A company makes ice cream and sells it to shops.

The ice cream is made in batches: a large quantity of one type and flavour of ice cream that is then split into smaller quantities for sale.

The company's owner has designed a relational database, `ICECREAM`, to store data about their ice cream and customers.

Some of the tables in the database are given. The database is not normalised.

`BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)`

`CUSTOMER(CustomerID, CompanyName, EmailAddress, TelephoneNumber)`

`SALE(SaleID, BatchID, CustomerID, Quantity, Date)`

(c) The table definition for BATCH is repeated here:

BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)

Sample data for the table BATCH is given:

BatchID	Type	Flavour	Size	SellingPrice	EndDate
KIV12	Plain	Vanilla	1	2.20	12/12/2024
RIC14	Plain	Chocolate	0.5	1.80	12/12/2024
TYL1	Non-dairy	Lemon	0.5	2.10	01/02/2025
FYV2	Non-dairy	Vanilla	0.25	1.50	02/02/2025
BIV13	Plain	Vanilla	1	2.20	02/02/2024

(ii) The table BATCH is not normalised.

Normalise the database table BATCH.

Write the table definitions for your new tables.

Identify any primary and foreign keys in your tables.

Do **not** change or include the tables CUSTOMER and SALE.

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Question 5 | Source: 9618_s24_qp_11 (Q6.a)

- 6 A company is developing a website that will allow users to create an account and then play a quiz every day. The data about the users and the quizzes are stored in a database.

A user must select a unique username and enter a valid email address to create an account. All users must be over the age of 16. A new quiz is given to the users every day. Each quiz is stored in its own text file.

The database stores the filename of each quiz and the date it can be played. The user gets a score for each quiz they complete, which is stored in the database. The scores are used to give each user a rating, for example Gold.

(a) Create a 3-table design for this database normalised to Third Normal Form (3NF).

Give your table design in the format:

TableName (PrimaryKey, Field1, Field2, ...)

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Question 6 | Source: 9618_s24_qp_12 (Q4.d)

- 4 An assessment board wants to store the marks students achieved in exams in a database named RECORDS.

Part of the database design includes these two tables:

EXAM(ExamID, Subject, Level, TotalMarks)

EXAM_QUESTION(ExamQuestionID, ExamID, QuestionNumber, Question, MaxMark)

- (d) The database also needs to store data about the students, the exams the students have taken and the marks the students achieved in each question of each exam.

Describe the additional tables that will need to be included in the database **and** explain how all the tables in the database will be linked.

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Question 7 | Source: 9618_s24_qp_13 (Q4.d)

- 4 A theatre wants to use a database to store data about the shows that are scheduled, their customers and the seats that the customers have booked.

In the theatre:

- Each show can take place on multiple dates.
- Some dates can have more than one performance.
- There are multiple rows of seats.
- Each seat can be individually booked by its row letter and seat number, for example row E seat 2.

Part of the database design includes these tables:

SHOW(ShowID, Title, Duration)
SEAT(SeatID, RowLetter, SeatNumber)
PERFORMANCE(PerformanceID, ShowID, ShowDate, StartTime)

- (d) Customers give their first name, last name and email address when they are making a booking. One booking can include multiple seats.

Describe the additional tables that will need to be included in the database **and** explain how these tables will be linked within the database.

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Question 8 | Source: 9618_w23_qp_12 (Q2.c)

- (c) The following database table is not normalised.

StudentName	DateOfBirth	TutorGroup	Subject	SubjectCode
Yuwei Chen	01/09/2004	SMH	English, Maths, Computer Science	EN, MA, CS
Claudia Raj	23/02/2005	JMB	Maths, Physics, Art	MA, PY, AR
Aamil Akram	24/01/2005	KMB	Art, Design, English language	AR, DE, EN
Areeba Faraz	21/12/2004	SMH	English language, Chemistry, Design	EN, CH, DE

Explain how to modify the table to put it into First Normal Form (1NF).

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Question 9 | Source: 9618_s23_qp_11 (Q2.b.ii)

2 An organisation uses a database to store data about the types of bird that people have seen.

(b) The database, `Birds`, stores information about the types of bird and the people who have seen them.

Data about each bird seen is stored with its location and data about the person who saw the bird.

Database `Birds` has the following tables:

```
BIRD_TYPE(BirdID, Name, Size)

BIRD_SEEN(SeenID, BirdID, Date, Location, PersonID)

PERSON(PersonID, FirstName, LastName, EmailAddress)
```

(ii) The database `Birds` has been normalised.

Draw **one** line from each Normal Form to the most appropriate definition.

Normal Form	Definition
First Normal Form (1NF)	All fields are fully dependent on the primary key.
Second Normal Form (2NF)	There are no repeating groups of attributes.
Third Normal Form (3NF)	There are no partial dependencies.

[1]

Question 10 | Source: 9618_s23_qp_13 (Q4.c)

4 A shop rents cars to customers. The shop uses a relational database to store information about the rentals.

(c) The car rental database is not normalised. The current database design is:

```
BOOKING (CarRegistration, StartDate, EndDate,  
         CarModel, CarColour, CustomerFirstName)
```

```
CUSTOMER (CustomerFirstName, CustomerLastName, EmailAddress,  
          TelephoneNumber)
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Write a normalised database design for this database.

All tables must be in Third Normal Form (3NF).

Use the field names given **and** underline the primary key fields.

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Question 11 | Source: 9618_w22_qp_11 (Q4.b)

- 4 A photographer creates a relational database to store data about photographs taken at birthday parties.

The database, PHOTOGRAPHS, stores details of the customer, the party, the photographs taken and the cameras used.

The photographer has several cameras that are used for taking the photographs at the parties.

Each camera has a specific lens type (for example, XY32Z) and lighting type (for example, F1672).

Data about each photograph is stored in the database including the party at which it was taken, the time it was taken and the camera used.

The database has these four tables:

```
CUSTOMER (CustomerID, FirstName, LastName, Telephone)
```

```
PARTY (PartyID, CustomerID, PartyDate, StartTime)
```

```
PHOTO_DATA (PhotoID, PartyID, TimeTaken, CameraID)
```

```
CAMERA_DATA (CameraID, LensType, LightingType)
```

- (b) The database is normalised and is in Third Normal Form (3NF).

Describe the characteristics of a database that is in Third Normal Form (3NF).

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Question 12 | Source: 9618_w22_qp_12 (Q5.a)

- 5 A relational database, GARDEN, has the following tables:

OWNER(OwnerID, FirstName, TelephoneNo, TreeID, TreePosition)

TREE(TreeID, ScientificName, MaxHeight, FastGrowing)

- (a) The database is **not** in Third Normal Form (3NF).

Explain how the database can be normalised to 3NF.

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Question 13 | Source: 9618_w21_qp_12 (Q6.a.i)

- 6 A shop sells plants to customers. The shop manager has a relational database to keep track of the sales.

The database, PLANTSALLES, has the following structure:

PLANT(PlantName, QuantityInStock, Cost)

CUSTOMER(CustomerID, FirstName, LastName, Address, Email)

PURCHASE(PurchaseID, CustomerID)

PURCHASE_ITEM(PurchaseID, PlantName, Quantity)

- (a) The database is normalised.

(i) The table lists the following three stages of normalisation:

- The first stage is from a database that is not normalised (0NF) to First Normal Form (1NF).
- The second stage is from 1NF to Second Normal Form (2NF).
- The third stage is from 2NF to Third Normal Form (3NF).

Tick (✓) **one** box in each row to identify the appropriate stage for each task.

Task	Normalisation stage		
	0NF to 1NF	1NF to 2NF	2NF to 3NF
Remove any partial key dependencies			
Remove any repeating groups of attributes			
Remove any non-key dependencies			

[2]

Question 14 | Source: 9618_s21_qp_11 (Q7.b.ii)

7 Bobby and Kim are discussing databases.

(b) Bobby has a shop that sells products to customers. His database will store data about his customers, their payment details, orders and the products he sells. Customers will have login details to access their accounts. The database will update customers' payment and login details without keeping any historical records.

(ii) Tick (✓) **one** box to identify the relationship that cannot be directly implemented in a normalised relational database.

Relationship	Tick (✓)
one-to-one	
one-to-many	
many-to-many	

[1]

Question 15 | Source: 9618_s21_qp_12 (Q1.b)

1 Raj owns houses that other people rent from him. He has a database that stores details about the people who rent houses, and the houses they rent. The database, HOUSE_RENTALS, has the following structure:

```
CUSTOMER(CustomerID, FirstName, LastName, DateOfBirth, Email)
HOUSE(HouseID, HouseNumber, Road, Town, Bedrooms, Bathrooms)
RENTAL(RentalID, CustomerID, HouseID, MonthlyCost, DepositPaid)
```

- (b) Tick (✓) **one** box to identify whether the database `HOUSE_RENTALS` is in Third Normal Form (3NF) or not in 3NF.
Justify your choice using one or more examples from the database `HOUSE_RENTALS`.

In 3NF	
Not in 3NF	

Justification

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MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q2.b)

2(b)	1 mark per bullet point, max 2 marks • There are no repeating groups of attributes • There are no many-to-many relationships • There are no partial key dependencies // no non-key dependencies // no transitive dependencies	2
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Mark Scheme for Question 2 | Source: 9618_s25_ms_11 (Q5.d)

5(d)	1 mark each to max 3 e.g. • Each order would only be able to have one item • or the database would not be normalised • it would not be in 1NF • due to repeated groups of attributes • in the ORDER table	3
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Mark Scheme for Question 3 | Source: 9618_s25_ms_12 (Q5.a)

5(a)	1 mark each to max 2 • To remove the many-to-many relationship • between CHARACTER and ITEM tables • To allow each character to have many items // to allow each item to be purchased for many characters • by creating a linking table • between the characters and the items purchased for each one	2
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Mark Scheme for Question 4 | Source: 9618_w24_ms_13 (Q4.c.ii)

4(c)(ii)	1 mark for each bullet point each (max 4) e.g. • Ice cream table with an appropriate name • ... containing type, flavour, size, selling price • ... with suitable primary key • ... foreign key identified in BATCH that links to the primary key in ICECREAM Example table definitions – not example answer: <pre>BATCH(<u>BatchID</u>, IceCreamID, EndDate) ICE_CREAM(<u>IceCreamID</u>, Type, Flavour, Size, SellingPrice)</pre>	4
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Mark Scheme for Question 5 | Source: 9618_s24_ms_11 (Q6.a)

6(a)	1 mark each: • User table with the username as the Primary Key • ... containing at least email address, date of birth / age and rating • Quiz table with Quiz ID or date or file name as the Primary Key. • ... containing at least the other field(s) not used as the PK • A joining table with an appropriate name including at least fields for user identification, quiz identification and score • ... with an appropriate Primary Key • ... and Foreign Keys matching the Primary Keys of the other two tables <pre>USER(<u>Username</u>, Email, DateOfBirth, Rating) QUIZ(<u>QuizID</u>, Date, Filename) USER_QUIZ(<u>Username</u>, <u>QuizID</u>, Score)</pre>	6
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Mark Scheme for Question 6 | Source: 9618_s24_ms_12 (Q4.d)

4(d)	1 mark each to max 5: - STUDENT table identified with suitable Primary Key - A linking table between STUDENT and EXAM with suitable Primary Key and appropriate name ... that includes the Primary Key of the STUDENT table as a Foreign Key to join with STUDENT ... and includes the Primary Key of the EXAM table as a Foreign Key to join with EXAM - A linking table between STUDENT and EXAM_QUESTION with suitable Primary Key and appropriate name ... that includes the Primary Key of Table 2 as a Foreign Key to join with Table 2 ... that stores the ExamQuestionID and the mark for that question	5
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Mark Scheme for Question 7 | Source: 9618_s24_ms_13 (Q4.d)

4(d)	1 mark each to max 5: e.g. - CUSTOMER table identified with suitable Primary Key and appropriate name ... and other suitable fields including name and email - BOOKING TABLE identified with suitable Primary Key and appropriate name ... that stores the Primary Key of the CUSTOMER table as a Foreign Key to join with CUSTOMER table ... and stores the Primary Key of the PERFORMANCE table as a Foreign Key to join with PERFORMANCE table - A linking table between Table 2 and SEAT with suitable Primary Key and appropriate name ... that includes the Primary Key of Table 2 as a Foreign Key to join with Table 2 ... that stores the SeatID.	5
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Mark Scheme for Question 8 | Source: 9618_w23_ms_12 (Q2.c)

2(c)	1 mark for each bullet point (max 4) - Identify repeating groups of attributes Subject and SubjectCode - Ensure each field is atomic ... StudentName should be split into e.g. FirstName and LastName - Identify the primary key for the table	4
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Mark Scheme for Question 9 | Source: 9618_s23_ms_11 (Q2.b.ii)

2(b)(ii)

1 mark for all 3 correct lines

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Normal Form

First Normal Form
(1NF)

Second Normal
Form (2NF)

Third Normal Form
(3NF)

Definition

All fields are fully
dependent on the primary
key.

There are no repeating
groups of attributes.

There are no partial
dependencies.

Mark Scheme for Question 10 | Source: 9618_s23_ms_13 (Q4.c)

4(c)	1 mark each - Only 3 tables with appropriate identifiers (i.e. one table for customer, one for booking and one for car) - Appropriate Primary key in each table underlined - Booking table includes Primary key from car and Primary key from customer as Foreign keys - All original fields are in correct tables Example answer: <pre>BOOKING(<u>BookingID</u>, CarRegistration, CustomerID, StartDate, EndDate) CAR(<u>CarRegistration</u>, CarModel, CarColour) CUSTOMER(<u>CustomerID</u>, CustomerFirstName, CustomerLastName, EmailAddress, TelephoneNumber)</pre>	4
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Mark Scheme for Question 11 | Source: 9618_w22_ms_11 (Q4.b)

4(b)	1 mark for each bullet point (max 3): - no repeating groups of attributes // data is atomic - no partial key dependencies - no non-key dependencies // no transitive dependencies	3
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Mark Scheme for Question 12 | Source: 9618_w22_ms_12 (Q5.a)

5(a)	1 mark for each bullet point (max 3): Solution 1: - removing the many-to-many relationship between Owner and Tree ... by removing TreeID and TreePosition from the Owner table ... and creating a linking table between Owner and Tree ... that contains OwnerID, TreeID and TreePosition ... (composite) primary key of the linking table should be OwnerID and TreeID // insert a named new primary key in the linking table Solution 2: - removing the many-to-many relationship between Owner and Tree - move TreePosition into TREE table ... put OwnerID into TREE table - create a new table with suitable name (for the species of tree) ... containing ScientificName, MaxHeight and FastGrowing ... with ScientificName as primary key // or another suitable primary key	3
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Mark Scheme for Question 13 | Source: 9618_w21_ms_12 (Q6.a.i)

6(a)(i)	1 mark for 1 tick in the correct place 2 marks for all 3 ticks correct			2	
	Task	Normalisation stage			
		0NF to 1NF	1NF to 2NF		2NF to 3NF
	Remove any partial key dependencies		✓		
	Remove any repeating groups of attributes	✓			
	Remove any non-key dependencies				✓

Mark Scheme for Question 14 | Source: 9618_s21_ms_11 (Q7.b.ii)

7(b)(ii)	1 mark	1
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Mark Scheme for Question 15 | Source: 9618_s21_ms_12 (Q1.b)

1(b)	1 mark per bullet point to max 2 - All fields in all tables are dependant fully on the PK and on no other fields - for example all fields in Customer table are fully dependent on CustomerID	2
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